

Patellar chondral defects: a review of a challenging entity

George Mouzopoulos · Carlo Borbon ·
Rainer Siebold

Received: 18 November 2010 / Accepted: 5 May 2011 / Published online: 24 May 2011
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Abstract

Purpose The aim of this study was to perform a review of the management of patellar cartilage defects, identify prognostic factors for better clinical outcomes, and propose a treatment algorithm of this challenging entity.

Methods We conducted a review of multiple databases, evaluating the clinical outcomes after patellar cartilage lesion treatment. Because of the heterogeneity of data, a meta-analysis could not be performed.

Results Twelve studies were identified for inclusion. Based on the clinical outcomes of these studies, a treatment algorithm was proposed. The clinical outcomes after patellar cartilage defects repair depend on the location and size of the articular defect, the age of the patient, the successful reconstruction of the patellar ridge, the repair of extensor mechanism malalignment, and the coexistence of chondromalacia.

Conclusion The optimal treatment for cartilaginous defects of the patella is still elusive. More prospective studies are needed, in order to identify which techniques are cost-effective especially on a long-term basis.

Level of evidence IV.

Keywords Patellar · Knee · Chondral · Cartilage · Defect · Lesion

Introduction

Cartilage defects of the patella are often misdiagnosed, due to subtle clinical symptoms and difficulties in localizing

and visualizing the retropatellar region on physical examination and plain radiographs, respectively [36].

Usually, these lesions are recognized accidentally during the treatment of another pathology [38]. As a matter of fact, articular cartilage defects normally do not heal spontaneously and may predispose the joint to the development of secondary osteoarthritis [4, 10]. Generally, there are no evidence-based methods for the treatment of cartilage defects in the knee [4].

Although a number of different treatment options exist, none of these may be identified as the golden standard [4, 57]. So it is important for the physicians to have objective criteria about the effectiveness of the treatment options of patellar chondral defects, as most decisions on funding are based on reliable data. Besides, only few studies exist in the literature to guide the surgeons in effectively managing these lesions.

The purpose of this review was to report all the available surgical options used for the treatment of patellar cartilage defects, present their effectiveness and propose a treatment algorithm, in order to assist surgeons in the treatment of this challenging entity. A literature search in PubMed, the Cochrane Central Register of Controlled Trials, MEDLINE, EMBASE, WHO International Clinical Trials Registry Platform and Current Control Trials, Cumulative Index for Nursing and Allied Health Literature and SPORTDiscus with no language restrictions was carried out (last search was carried out on September 2010). A publication period from 1960 to 2010 was selected. All articles were manually reviewed by the authors of this study, and references were also reviewed for possible inclusion. The search terms were knee, patellar, chondral, cartilage, lesion, and defect. A total of 738 abstracts were screened by the authors. Of these, only 12 studies could be extracted that met the following criteria: treating

G. Mouzopoulos (✉) · C. Borbon · R. Siebold
Center for Knee and Foot Surgery, ATOS Clinic,
Bismarckstr. 9-15, 69115 Heidelberg, Germany
e-mail: gmouzopoulos@yahoo.gr

exclusively cases of patellar cartilage defect and stating clearly the clinical outcomes, published between 1960 and 2010, available in the above databases and being online accessible. Only papers evaluating patellar cartilage defect surgery without any additional procedures were considered as any additional procedure would influence the clinical outcomes. Because of the heterogeneity of the studies found, meta-analysis could not be performed.

Herein, we discuss the clinical presentation, historical features, and diagnostic evaluation and emphasize on the available treatment options and prognosis of patellar cartilage lesions. Also, a treatment algorithm based on the specific indications of each treatment option is proposed for the adequate treatment of patellar cartilage lesions.

Causes

Patellar chondral lesions are often associated with abnormal patellofemoral stress, such as lateral compression or excessive lateral position of the patella on the trochlea [28, 45]. Such disorders lead to altered articular congruence between patella and trochlea that can progress to severe cartilage damage of one or both patellar facets. However, patella cartilage lesions may happen acutely due to sport trauma or traffic accidents causing patella fracture or dislocation [51]. Other rare causes include osteochondritis dissecans and idiopathic chondromalacia [45].

Clinical symptoms

Patellar cartilage defects may cause pain, swelling, clicking, and instability, impairing daily activities [9]. Moreover, these chondral lesions may lead to early degenerative changes [45]. On the other hand, not rarely these articular cartilage injuries are presented without symptoms and are recognized accidentally on MRI or during arthroscopy procedure [10].

Incidence—classification

Cartilage defects of the knee are commonly seen during routine arthroscopy [59]. A review of 31,516 knee arthroscopies reported a 63% prevalence of chondral lesions, with the patella being the most common location [10, 15]. A topographic study of 105 patients with anterior knee pain documented that the most frequent site of cartilage lesion (72%) is a roughly elliptical area with the major diameter parallel to the transverse axis of the patella, not affecting the upper and lower thirds of the patella [45]. The lateral and medial facets are affected in 7 and 21% of

cases, respectively [45]. Regarding the depth and size, the majority of patella cartilage defects are Outerbridge grade III–IV chondromalacia [44]. The classification systems referred to the location and depth size of the patellar cartilage damage are listed in Table 1 [17, 44].

Diagnosis

MRI is a useful tool in detecting patellar chondral lesions [13]. Grade I lesions are usually seen on “proton density” and T2 MRI images as round areas of low signal intensity within the cartilage. Grade II lesions often appear as zones of low signal surrounded by high signal within the cartilage on T2 images. Grade III lesions are easily recognized as superficial or deep defects of the cartilage surface and appear bright when filled by synovial fluid (high signal intensity) on T2-weighted images. Thinning and irregularity of the contour of the cartilage are easily diagnosed in degenerative arthritis grade IV lesions. CT with contrast is a valuable but relatively invasive tool in the assessment of patellar cartilage, but appears less reliable than MR imaging in lesions smaller than 3 mm [13]. Arthroscopy is considered as the golden standard in diagnosing cartilage defects by direct vision [10]. As a matter of fact, compared with arthroscopy, MR imaging has a sensitivity of 100%, a specificity of 50% and a relatively high accuracy of 81.5% in evaluating patellar cartilage [23]. Besides, MRI allows staging of chondral defects and might detect very early cartilage lesions not diagnosed in arthroscopy [23].

Treatment options

Nonsurgical treatment with chondroprotective medication, nonsteroid anti-inflammatory drugs, and physiotherapy is well established for low-grade, superficial chondral defects [33].

Surgical treatment has been suggested for deep chondral defects. The indications of surgical repair of patellar chondral defects are listed in Table 2. Although surgical approaches to the patellar cartilage lesions have received less attention, probably due to the poor range of cartilage repair options available until not long ago, recently different operative procedures are applied for patellar articular cartilage repair (Table 3).

Marrow-stimulation procedures

Typically, symptomatic cartilage injuries are treated with marrow-stimulation procedures as a first-line treatment. Marrow-stimulation procedures, such as microfracture,

Table 1 Classification of patellar chondral defects

Fulkerson classification (location)
Type I : articular injury of the inferior pole of the patella
Type II: articular injury of the lateral facet of the patella
Type III: articular injury of the medial facet of the patella
Type IV: articular injury of the proximal pole (type IVa) or a panpatellar injury (type IVb)
Outerbridge classification (depth size)
Grade 0: normal cartilage
Grade I: cartilage with softening and swelling
Grade II: a partial thickness defect with fissures on the surface that do not reach subchondral bone or exceed 1.5 cm in diameter
Grade III: fissuring to the level of subchondral bone in an area with a diameter more than 1.5 cm
Grade IV: exposed subchondral bone

Table 2 Indications of surgical repair of patellar chondral defects

Persistent clinical symptoms such as severe anterior knee pain, recurrent swelling, or clicking
Radiologic and arthroscopic proof of an osteochondral defect with an unstable fragment
Patellofemoral osteoarthritis development
Failed conservative treatment >6 months

Table 3 Types of surgical repair of retropatellar chondral defects

Marrow-stimulation techniques (microfracture, drilling, abrasion)
Microfracture combined with autologous matrix-induced chondrogenesis (AMIC)
Realignment of extensor mechanism
Osteochondral transplantation (mosaic plasty)
Periosteal transplantation
Chondrocyte implantation
First generation with the use of periosteum or synthetic membrane (MACI) as scaffold
Second-generation autologous chondrocyte implantation (ACI)
Osteochondral allograft
Patellofemoral arthroplasty

drilling and abrasion, have been proposed as an easy, rapid, and inexpensive way to restore the articular-damaged cartilage [54].

Microfracture is performed according to the technique described by Steadman et al. [53]. An initial thorough diagnostic examination of the knee is performed. After identifying the full-thickness chondral lesion, the unstable cartilage, including cartilage loosely attached to the surrounding rim, is removed, using a shaver and/or a handheld angled curette. Once the exposed subchondral bone plate is thoroughly debrided, multiple v-shaped holes using a

Steadman arthroscopic pin or specially bent awl are made. The holes are placed perpendicular to the joint surface, approximately 3–4 mm apart and about 2–4 mm deep, with care taken not to harm the subchondral plate between the holes. Once the holes are completed, the irrigation fluid pump pressure is lowered to visualize the release of fat droplets and blood from the microfracture holes into the knee. This method can be safely applied in small isolated cartilage defects (1–3 cm²) in young, active patients [54]. Similarly, the drilling technique is done by making multiple holes, in the same manner [48]. Also, the abrasive arthroplasty is performed with an arthroscopic shaver system [54]. The subchondral sclerotic bone is treated with a milling head, and the procedure is complete when a bleeding out of the bone is apparent.

All marrow-stimulation methods are based on the penetration of the subchondral bone plate at the bottom of the cartilage defect. The outflowing bone marrow blood contains the pluripotent stem cells (hMSC) that are stabilized by the clot formation in the defect [9]. The number of these highly proliferative stem cells during this procedure is very low, and the concentration in the bone marrow is age dependent [7]. The hMSC that are able to differentiate into fibrochondrocytes result in fibrocartilage repair with varying amounts of type I, II, and III collagen [53, 58]. The replacement tissue after opening of the subchondral space resembles fibrocartilage or dense connective tissue more than hyaline cartilage. This newly formed fibrocartilage possesses a smaller amount of proteoglycans than that found in hyaline cartilage [54]. Also, these regenerates are often unstable and have a low mechanical stress resistance [53]. So the patient's benefit is often poor and of a short duration.

In recent years, older techniques such as Pridie drilling or abrasion have been increasingly replaced by the microfracture technique [52]. However, despite its widespread use, there are no published reports on microfracture or drilling alone for patellar cartilage defects. In one study, the patellar abrasive arthroplasty produced a significant increase in the Lysholm score, but no reduction of the retropatellar pain was mentioned [51].

Microfracture combined with autologous matrix-induced chondrogenesis (AMIC)

The AMIC (autologous matrix-induced chondrogenesis) technique was developed to manage larger defects (>3 cm²) by application of a collagen type III/I membrane. AMIC is particularly indicated for treating clinical symptomatic retropatellar cartilage defects more than 1 cm and less than 8 cm [54]. This technique has become a real

alternative, since these defects cannot be effectively treated with standard microfracturing alone.

Using minimal medial approach, the destroyed and unstable cartilage is removed using a scalpel, curette and spoon, until a well-contained defect results [3]. Then, microfracture technique is performed as previously mentioned. An imprint of the defect is taken using an aluminum template, and the Chondro-Gide collagen membrane is cut slightly smaller than this template. Application of ringer salt solution to the membrane will later result in an increase in the size by approximately 10% [54]. The membrane can be placed precisely with the rough side to the preserved bone plate using fibrin glue. To prevent delamination, laying the membrane edges over the rim of the cartilage should be avoided [3]. With this autologous regenerative approach, the stem cells (hMSC) available in the bone marrow are brought to the surface by microfracturing and so become available for cartilage repair [32]. The collagen matrix serves as a natural scaffold for cell binding and should stimulate differentiation processes [16].

The AMIC procedure provides two major advantages: it is a one-step procedure with no need of cartilage harvesting and it is cost-effective with no need of *in vitro* cell expansion [3]. The limitations of the technique are summarized in a technically demanding and time-consuming procedure [54]. Besides, the graft is secured to the surrounding native cartilage by sutures and may further damage the uninjured cartilage [12].

Gille et al. [19] treated 9 patients with this method and after 2 years of follow-up reported a significant increase in values of Lysholm and IKDC scores. Also interestingly, no hypertrophy of repair tissue was seen in MRI follow-up studies at the patella. The authors concluded that AMIC enhance the results of microfracture technique in terms of quality and durability.

Recently, Dhollander et al. introduced the AMIC plus technique that consisted of the AMIC procedure combined with platelet plasma. The authors reported good clinical outcomes in five patients after 2 years of follow-up [11].

Realignment procedures of extensor mechanism

Additionally, treatment options such as anteromedialization of tibia tubercle (AMZ), medial patellofemoral ligament reconstruction, and lateral release, with or without microfracture, could be considered as well as treatments for structural abnormalities of the extensor mechanism in patients with chondral defects in the patellofemoral joint [6, 47].

Symptomatic defects of the patella located distally and laterally are often treated initially with an AMZ with or without a microfracture procedure, as these have

historically managed well with AMZ alone as reviewed by Pidioriano et al. [47]. When defects involve the central or proximal patella, especially when more medially located, modifications to the AMZ should be considered. These include specific attention to avoid over-medialization, which might increase medial patellofemoral contact pressures [47]. The tibia tubercle osteotomy is done with a 60 slope for medial/central patellar lesions (more anteriorization of the tibial tuberosity) [47]. For lateral patellar lesions, the slope is decreased to 45 (increasing medialization relative to anteriorization) [47]. Then, tuberosity is repositioned and fixed with two 4.5-mm interfragmentary screws in the new position with standard AO technique.

Pidioriano et al. [47] in a retrospective study of 37 patients, suffered from patellar articular defects and treated with AMZ, tried to correlate the location of the patellar lesions with overall outcome. After a follow-up of almost 4 years, they reported that patients with distal and lateral facet lesions are more likely to have good or excellent result and experience less pain than patients with medial, proximal, or diffuse lesions. The conclusion of their study was that completely normal knee function after this procedure may not occur, but the patients may experience improvement in pain, stability, and activities of daily living and possibly an increase in the level of sports participation. Also they mentioned the importance of concomitant central trochlear lesion. As a matter of fact, six of seven patients with central trochlear lesions had poor outcome. Finally, they underlined that the contraindications of this procedure should be medial, proximal, or diffuse patellar lesions and concomitant central trochlear defects.

Osteochondral transplantation

The principle of this method is to harvest a bone cartilage graft from the posterior part of the femoral condyle and implant it into the site of the defect that is localized in a part of the joint that is more important for load transmission [34, 49].

A dorsal approach through the popliteal fossa is performed to harvest the grafts from the posterior part of the femoral condyles. When one graft is required, the graft is harvested from the contralateral femoral condyle. Otherwise, if two grafts are needed, they are obtained from both the femoral condyles. For graft harvesting, Laprell and Petersen used a diamond bone-cutting system and a special extractor [34]. Afterward, an anteromedial arthrotomy is performed. The implantation site for the graft cylinder is prepared with the same instrument as for graft harvesting, but the size of the diamond bone cutter used for excision of the defect is chosen to be 1 mm smaller than the graft diameter to allow press-fit implantation. Press-fit implantation is then performed under manual pressure.

No large series that used osteochondral transplantation are mentioned in the literature. Laprell and Petersen [34] treated 35 patients with knee chondral defects, three of those located in the retropatellar region, but the results regarding the patellar lesions were not specified. Rose et al. [49] treated one patient with this method and reported good results after 6 months of follow-up.

A variant of osteochondral transplantation is the so-called mosaic plasty. The basic idea of this procedure was that the use of small-sized multiple cylindrical grafts harvested from the relatively less weight-bearing areas rather than a single large block graft would allow more tissue to be transplanted while preserving donor site integrity, and the mosaic-like implanting fashion would permit progressive contouring of the new surface [24].

Autologous osteochondral mosaic plasty was used to treat small- and medium-sized (1–3 cm²) focal chondral and osteochondral defects of the weight-bearing surfaces of the patellofemoral joint [30]. Fifty years of age constitutes the recommended upper limit for this procedure, as the repair capacity is diminishing with the increasing age [24].

The main concern is whether the donor site easily progresses to osteoarthritis development. However, there is no evidence that graft harvest results in further degenerative changes. Sometimes the patients complain of knee pain, but this could be avoided if the grafts are small and peripherally located [5].

Hangody et al. [24] reported excellent results in 74% of patients with patellar defects who were treated with mosaic plasty after 4 years of follow-up. Despite these promising results, Bentley et al. [5] reported failure in all five patients undergoing mosaic plasty for articular cartilage lesions of the patella.

Periosteal transplantation

The pluripotent mesenchymal cells in the cambium layer of the periosteum have the capacity to differentiate into hyaline-like articular cartilage containing type II collagen, as has been shown in both experimental and clinical studies [27, 50].

The chondral lesion is removed, the sclerotic subchondral bone is excised, and multiple drilling holes are made through the remaining subchondral bone into the cancellous bone [27]. Drill holes through the patella are placed in the corners and at the sides of the articular defect, near the borders of the surrounding healthy cartilage. The periosteum is taken from the proximal tibia metaphysis, ventral of the pes anserinus, and is anchored to the underlying bed with the cambium layer (inner layer) turned inward. The periosteal transplant is anchored to the bottom of the cartilage defect with the use of through and through sutures

tied on the dorsal side of the patella or sutured the transplant to the surrounding cartilage. In order to enhance the fixation, fibrin glue (Tisseel) could be injected under the transplant.

Regarding the technical aspects of the procedure, two major points are very crucial. First, the sclerotic bone plate will constitute an area with less compliance with loads, so the sclerotic part of the subchondral bone must always be removed and the plate perforated [43]. Therefore, there is a possibility for growth factors and mesenchymal stem cells from the subchondral bone to be differentiated in chondrocytes beneath the periosteum. It is well known that growth factors such as transforming growth factor- β and basic fibroblastic growth factor are involved in the process of regeneration of cartilage [29]. Mesenchymal stem cells are also capable of forming new hyaline-like cartilage [58].

Secondly, the anchoring technique is very important to prevent loosening of the transplanted periosteum during the rehabilitation period. The articular surfaces sustain high loads, so wearing or shearing effects of the transplant may occur during knee movement, especially if its fixation is poor [31].

In the study by Hoikka et al. [27], 13 patients were included (3 had chondromalacia and 10 suffered traumatic injuries), and in 10 patients, immobilization in a cast was used postoperatively. The results at follow-up for a mean of 48 months postoperatively were unsatisfactory (five graded as fair or poor). Korkala and Kuokkanen [31] have reported the results from seven patients (three had chondromalacia and four had traumatic injuries) managed with periosteal transplantation. Postoperatively, two patients were immobilized in a cast, four patients were treated with CPM, and one patient was managed with controlled motion. From that report, it is not possible to identify the results of each case, but three of the seven patients had an excellent or good outcome at follow-up of 1.5–4 years postoperatively. Basically, using immobilization in a cast after periosteal transplantation is biologically incorrect, because experimental studies have clearly shown that rigorous immediate postoperative motion plays a central role in the differentiation of the mesenchymal cells in the periosteal cambium layer into chondrocytes and finally hyaline cartilage [35, 43].

One study reported 96% good to excellent results in isolated patellar cartilage defects treated by periosteal transplantation, using a particular periosteal suture, associated with continuous passive motion in the immediate postoperative period [35]. Alfredson and Lorentzon [1] found that autologous periosteal transplantation in patients with disabling knee pain from a full-thickness cartilage defect of the patella is associated with promising results and with a high percentage of patients able to return back to pain-free walking and to previous daily activities.

Postoperative MRI examinations showed filling of the defect with a tissue that resembled the surrounding cartilage. In addition, biopsies revealed hyaline-like cartilage. Also, the authors concluded that the patients who were treated with CPM in the immediate postoperative period after periosteal transplantation to deep cartilage defects of the patella have a better outcome compared with the patients who were treated with active motion in the same period. A possible explanation might be that the compression forces on the surface of the patella, due to muscle activity, are much higher during active motion. This might impair the cartilage regeneration. Furthermore, in this study, it was remarkable that in the group of patients who were treated with CPM postoperatively and followed for a mean of 51 months, only 4 of 38 patients (11%) showed signs of deterioration with time. All these four patients suffered from chondromalacia (no known trauma). Therefore, the authors suggested not using periosteal transplantation in cases associated with chondromalacia of the patella. On the other hand, patients who have severe pain from cartilage injuries and who have failed conventional treatment should be considered for periosteal transplantation. Spahn and Kirschbaum [51] also treated 15 patients with periosteal transplantation. After 3 years of follow-up, they reported a 58.3% increase in Lysholm score and a 64.7% decrease in pain experienced according to visual analog scale (VAS).

Autologous chondrocyte implantation

In recent years, autologous chondrocyte implantation (ACI) has emerged as a potential therapeutic option. Use of ACI in clinical practice has involved the application of tissue-engineering techniques with a cell-based therapy, which allows the regeneration of damaged cartilage. Some studies suggest the durability of this treatment, especially at long-term follow-up, because of its ability to produce hyaline-like cartilage that is mechanically and functionally stable, and it allows integration with the adjacent articular surface [46].

The first-generation technique, introduced by Brittberg et al. [8], is complex, owing to the difficulty in handling the liquid chondrocyte solution. The main disadvantages of this technique include the requirement for harvesting periosteal tissue and the need for meticulous sewing of the periosteal patch over the defect to ensure a “water-tight” closure that prevents spillage of the chondrocytes and the frequent occurrence of periosteal hypertrophy [26]. As a matter of fact, Brittberg et al.’s [8] early results achieved only 29% good to excellent outcomes.

To address these problems, second-generation ACI has been developed, and biodegradable polymers as temporary

scaffolds for the *in vitro* growth of living cells, and their subsequent transplantation onto the defect, have become widely used. One of the most commonly used scaffolds for ACI is entirely based on the benzylic ester of hyaluronic acid and consists of a network of 20- μ m-thick fibers with interstices of variable sizes [37]. This scaffold has been demonstrated to be a good physical support to allow cell–cell contacts, cluster formation, and extracellular matrix deposition.

The whole procedure is performed in two stages [39]. First, cartilage biopsy specimens for chondrocyte harvest are obtained from the minimally weight-bearing portions of the knee (intercondylar notch), and the dimensions of the lesion are documented to estimate the size of the graft required. Afterward, the biopsy material is sent to the laboratory for *in vitro* isolation and expansion of autologous chondrocytes for a duration of 3 weeks. Subsequently, the cells are seeded onto the scaffold for 1 week to create the tissue-engineered product Hyalograft C [39]. On this scaffold, the cells are able to redifferentiate and retain a chondrocytic phenotype even after a long period of *in vitro* expansion in monolayer culture. At second stage, implantation is performed by mini-arthrotomy approach (lateral or medial parapatellar arthrotomy incision). The nonviable chondral tissue is removed with the use of a curette, the graft is adequately templated, and the properly sized graft is transplanted [39].

This second-generation technique with a hyaluronan-based scaffold eliminates the need for obtaining periosteal tissue from patients, circumvents the delicate step of handling a liquid chondrocyte culture, and in the majority of cases does not require additional fixation to attach the graft to the defect, thereby simplifying the procedure and decreasing the operative time [39]. The features of this device have also permitted the development of an arthroscopic surgical technique for the implantation of autologous chondrocytes on a hyaluronic acid support with the aim of reducing patient morbidity, surgical time, and recovery and complications related to open surgery [37]. The most important theoretical advantage of this method is that it can restore the integrity of the damaged area with cartilage that has hyaline-like properties, whereas all the marrow-stimulation techniques provide fibrocartilaginous repair tissue that is likely to be less mechanically stable [39].

Peterson et al. [46] obtained 57.8% good to excellent results at 4.2 years by associating, where necessary, an operative realignment of the extensor mechanism. Minas and Bryant [39] mentioned improvement in clinical scores after 3.86 years of follow-up. Gobbi et al. using a hyaluronic acid synthetic scaffold obtained almost 91% A and B results according to the IKDC knee ligament standard evaluation at 2 years [21]. In another study, Gobbi et al.

[20] performed a second-look arthroscopy almost 1 year after the initial procedure, in 6 patients with patellar defects treated with ACI. They reported cartilage repair assessment score near to normal in all cases. Lorentzon et al. [35] performed a second-look arthroscopy in 7 patients after periosteal transplantation and found that the transplanted area was covered with a thin layer that had the appearance of periosteum. The surface under the periosteum had a cartilage-like appearance macroscopically, and it was difficult to see the borders between the transplanted area and the surrounding cartilage. The transplanted area and the surrounding cartilage had a similar stiffness when probed [35].

Another variant of first-generation ACI is the so-called matrix autologous condrocyte implantation (MACI), which instead of periosteal flap uses a synthetic membrane as a scaffold. In a retrospective cohort study, Niemeyer et al. [42] compared the outcomes of periosteal patch-covered conventional ACI, collagen membrane-covered ACI, and matrix-associated ACI in treating 70 patients with retropatellar cartilage defects. The authors reported good to excellent result in approximately 70% of patients and found larger lesions, lesions on the medial patellar facet, and diffuse lesions to have a poorer prognosis than smaller lesions and lesions on the lateral patellar facet. The clinical outcomes of the three surgical techniques used were no significantly different. Bentley et al. [5] have shown that ACI is far superior to mosaic plasty in repairing patellar cartilage lesions. However, there is no agreement between the effective superiority of second-generation ACI and microfracture [42].

The high cost of cell culture for ACI and the cost of the membrane itself for MACI are the main disadvantages of both techniques. Their actual cost-effectiveness compared with more traditional techniques is therefore still unclear.

Osteochondral allografts transplantation

Fresh osteochondral allografts have recently been suggested for the treatment of focal or large chondral and osteochondral defects of the knee. Although this technique requires only a single procedure for implantation, the risk of infection transmission still remains [55].

However, no large series exist in the literature regarding the efficacy of this technique in the treatment of patellar lesions.

Prognosis

Articular cartilage lesions, with their inherent limited healing potential, remain a challenging problem for

orthopedic surgeons. The clinical studies found in the literature, which report the clinical outcomes after patella cartilage repair, are listed in Table 4. However, it is well documented that patellar cartilage defects are associated with worse clinical outcomes than the defects of femoral condyles [21]. Besides, studies have shown that the clinical outcomes after patellar cartilage repair are deteriorated with time [1, 35]. Specific biomechanical properties such as increase in shearing forces in the femoropatellar compartment after excessive debridement and malalignment of the extensor mechanism that has not been corrected may explain these observations [21, 39]. Also, previous knee surgeries, the location and size of patellar defect, and age are important factors that affect the final outcome [21, 37, 39].

The clinical importance of patellar ridge reconstruction

The inferior results commonly seen after treating retropatellar chondral defects lie in the specific biomechanical properties of the femoropatellar compartment of the knee joint [39]. The forces arising in the anterior compartment during stair-climbing are in excess of seven times the body weight, and there are greater shearing forces there than in the medial and lateral compartments [48]. As a matter of fact, these shearing forces are not favorable to the formation of cartilage regeneration, precluding adequate cartilage repair [56]. When the forces arise at the femoropatellar joint compartment, the patellar ridge is thought to have a significant meshing function in guiding the patella in the sliding bearing [46]. It stabilizes and centers the patella when the flexing movement is started and in internal an external rotation as well and reduces the concomitant gliding movements in the coronary plane [2]. In this way, the patellar ridge contributes directly to the reduction of shearing forces in the anterior knee joint compartment.

The anatomic reconstruction of the patella ridge is crucial during patellar defects repair. The height of the patellar ridge is dependent not so much on the subchondral bone as on the thickness of the cartilaginous tissue covering the subchondral bone [56]. Thus, flattening of the patellar ridge due to excessive cartilage debridement extending beyond the ridge seems to be associated with poorer treatment outcomes [39]. Minas and Bryant [39] suggested primary fixation of the periosteal flap by the classic Brittberg technique in the region of the ridge, so as to set the appropriate ridge height for the regeneration product (so-called tent-shaped suture technique). But it is not clear how long the sutured periosteal flap will remain in the area of the defect, preserving the adequate shape of patellar ridge. Besides, since the use of the periosteal flap is

Table 4 Studies reported clinical outcomes after retropatellar chondral defects repair

Author	Study	Number of patients/ treatment	Follow-up (years)	Defect size (cm ²)	Clinical outcome
Gigante et al. [18]	Prospective	16/MACI + REM	3	4 (3–9)	Improvement in clinical scores Modified Cincinnati Rating Scale: 200% Lysholm: 57.3% Tegner: 300% Kujala: 63.7% 93% of patients satisfied
Nho et al. (2008)	Prospective	22/MOSAIC PLASTY + REM	2.39	1.65	Improvement in clinical scores IKDC: 57.62% SF-36 : 24%
Niemeyer et al. [42]	–	11/ACI	3.46	(3–8)	Lysholm: 75, IKDC: 60
		13/ACI + RIDGE DEBR	3.92	(3–8)	Lysholm: 60, IKDC: 58
Gobbi et al. [21]	Prospective	22/ACI	2	4.7 (0.8–12)	According to IKDC score: 40.9% normal, 50% near normal 9.1% abnormal
Minas and Bryant [39]	–	8/ACI + REM	3.86	4.34	Marked improvement in pain relief Improvement in clinical scores: WOMAC: 38.53%, SF-36: 21.98% Modified Cincinnati Score: 41.32% KSS function score: 43% KSS knee score: 52.51%
Spahn and Kirschbanm [51]	–	20/ABRASION	3.2	–	Lysholm: 42.5, Tegner: 2.7
		15/PERIOSTEAL TR	3.1	–	Lysholm: 67.6, Tegner: 4.9
Bentley et al. [5]	Prospective randomized	20/ACI	1	–	According to Cincinnati Rating Scale: 25% excellent, 60% good, 15% fair
		5/MOSAIC PLASTY	1	–	60% good, 40% poor
Peterson et al. [46]	Retrospective	19/ACI + REM	4.2	4.7	According to Brittberg Clinical Score: 31.5% excellent, 26.3% good, 31.5% fair, 10.7% poor
Alfredson and Lorentzon [1]	–	38/PERIOSTEAL TR + CPM	4.25	(0.75–20)	76% excellent or good results
		19/PERIOSTEAL TR + AM	1.75	(0.75–20)	52% excellent or good results
Lorentzon et al. [35]	–	26/ACI	3.5	6.2 (0.75–20)	According to Brittberg Clinical Score: 65.38% excellent, 30.76% good 3.85% poor
Pidoriano et al. [47]	Retrospective	37/REM	3.9	–	Excellent or good results according to Lysholm score: in 60% of distal lesions in 54% of lateral facet lesions in 33% of medial facet lesions in 20% of proximal or panpatellar lesions
Hoikka et al. [27]	–	13/PERIOSTEAL TR	3.8	–	61.5% good, 30.7% fair, 7.8% poor

REM realignment of extensor mechanism, DEBR debridement, TR transplantation, CPM continuous passive motion, AM active motion

associated with high incidence of cartilage hypertrophy, use of a collagen membrane has been suggested, because it helps to contain the original height in the region of the patellar ridge [22]. However, this theoretical advantage

must certainly be subject to some doubt, since the primary stability necessary for the maintenance of the thickness of cartilage in the postoperative phase is not provided by most of the collagen membrane available. The double eye

technique has been described recently where no debridement is carried out in the ridge region when the cartilage has become soft, in order to preserve the patellar ridge [41].

In their study, Niemeyer et al. [42] reported better results in patients treated with ACI and preserving the patellar ridge compared with patients treated with ACI as well, but with excessive debridement and flattening the ridge.

The necessity of repairing extensor mechanism malalignment

Deciding when to perform the alignment of extensor mechanism is complex and based on multiple factors, including the region of patellofemoral chondrosis, the initial position of the patella relative to the trochlea, and the position of the tibial tuberosity relative to the trochlear groove midline [47].

As a matter of fact, persistent subluxation and tilt lead to mechanical overload with subsequent progressive chondral wear. Minas and Bryant [39] recommended a treatment algorithm of patellar chondral defect associated with the malalignment of the extensor mechanism. Arthroscopic lateral release is indicated for isolated patellar tilt without subluxation of the patella when the lateral retinacular structures of the patella are contracted and the mobility of the patella is limited. However, arthroscopic lateral release should not be done in case of subluxation associated with patellar tilt. If the mechanics are abnormal and there are no chondral changes, then medialization of the tibial tubercle done with a lateral release is highly effective without anterior translation. When the chondral changes to the patella are type I and type II (according to Fulkerson's classification), anteromedialization tibial tubercle osteotomy is successful in unloading the damaged articular surface with a patellar lateral release and possible vastus medialis obliquus advancement as needed. When maltracking is associated with type III or type IV patellar lesions or the trochlea is additionally involved, cartilage repair by ACI is considered to be effective [39].

Only few studies have addressed the effectiveness of combined distal realignment associated with cartilage repair in isolated patellar cartilage lesions. In a recent study, Henderson et al. [25] compared two groups of 22 patients: one with patellofemoral malalignment and patellar cartilage defects and the other with patellar cartilage defects only. The two groups were treated with ACI combined with distal or proximal realignment or with ACI alone, respectively. The authors reported 86% good to excellent outcomes in the first group versus 55% in the second group. They ascribed the difference to unloading

due to the distal osteotomy or to incorrect preoperative evaluation of the extent of malalignment in the patients of the second group and concluded that patellofemoral joint unloading with a realignment procedure is desirable to maximize the clinical outcome of ACI, even when no tracking anomalies are identified clinically. Their reoperation rate was 52% (10/22 and 13/22 patients, respectively). In another study, Farr considered 38 patients with patellar and/or trochlear cartilage lesions, of whom 28 underwent distal realignment prior to or simultaneously with ACI [14]. Thirteen of the 38 patients had isolated patellar cartilage lesions, but no data were provided as to the type of malalignment treated, and patellofemoral congruence was assessed only by a lateral radiograph and a Merchant axial view, without specification of the amount of subluxation. Overall, 25 of Farr's 38 patients required additional surgery, 18 for procedures other than screw removal [14].

Nonetheless, their median Modified Cincinnati scores ranged from 3 to 6 and the Lysholm score from 56 to 86. Gigante et al. treated 16 patients with MACI and distal realignment and after 3 years of follow-up reported excellent or good results in 93% of the patients [18].

Previous cartilage repair

Minas et al. [40] recently suggested that microfracture is associated with thickening of the subchondral bone, osseous overgrowth, and formation of cysts and may compromise subsequent revision surgery (such as ACI) due to damage to the subchondral plate. Interestingly, they found that defects that had prior treatment affecting the subchondral bone failed at a rate 3 times that of nontreated defects.

Location and size of articular defect

Distal and lateral facet lesions are more likely to have good or excellent result and experience less pain than patients with medial, proximal, or diffuse lesions [47]. Medial lesions are often associated with trochlea defects and affect the patellar ridge, impairing the clinical outcome [39].

In literature, some studies have reported no association between defect size and the clinical outcome, while other studies had reported a significant association [30, 42].

Age of the patient

Younger patients are associated with better clinical results, as the repair capacity is diminishing with the increasing

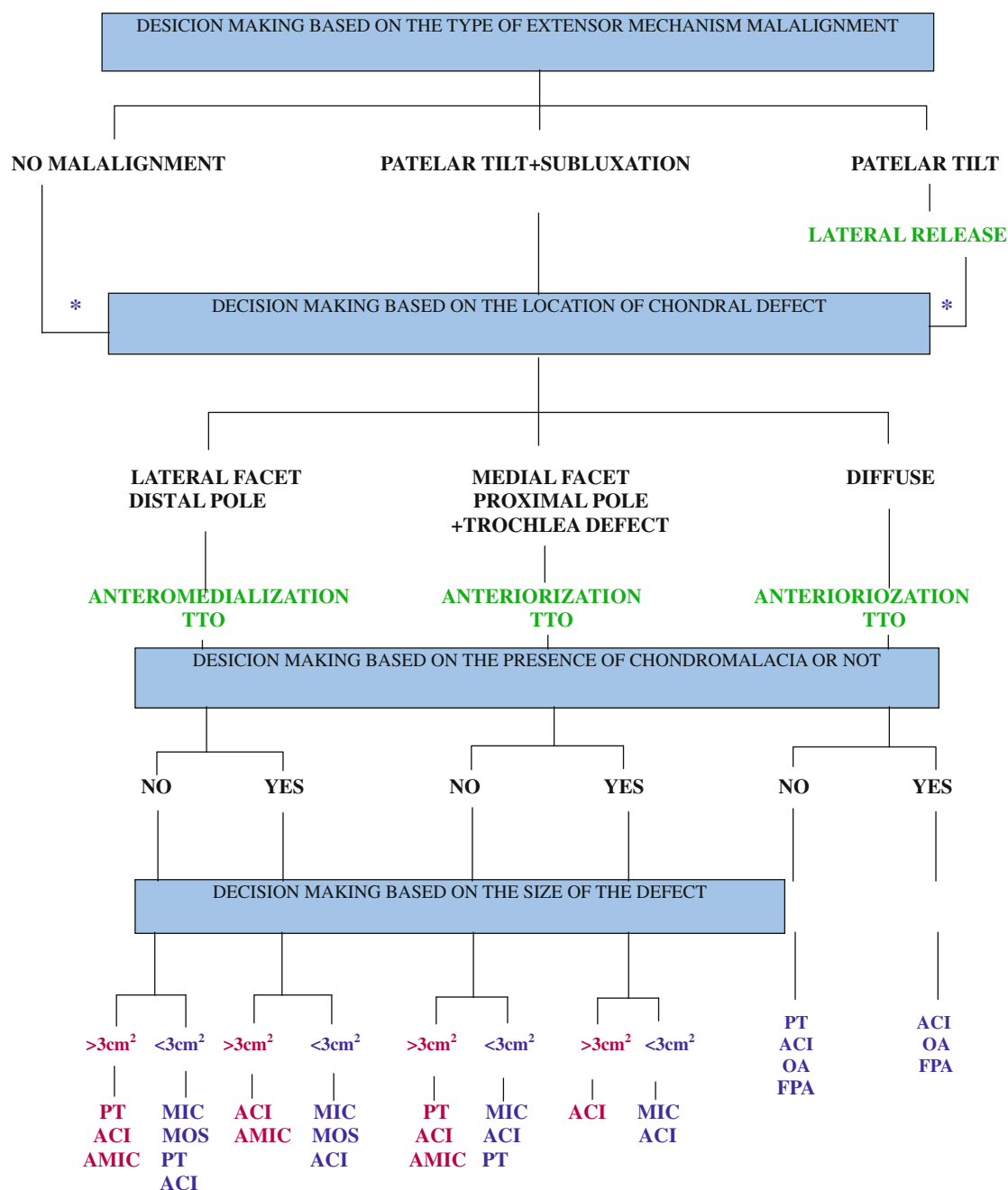


Fig. 1 Treatment algorithm of symptomatic patellar chondral defects. *Consider only the chondral defect treatment. *TTO* tibial tuberosity osteotomy, *PT* periosteal transplantation, *ACI* autologous

chondrocyte implantation, *AMIC* autologous matrix-induced chondrogenesis, *MOS* mosaic plasty, *OA* osteochondral allograft, *FPA* femoropatellar arthroplasty

age. According to Hangody et al., 50 years of age constitutes the recommended upper limit for performing osteochondral transplantation [24].

Suggested treatment algorithm

Taking into consideration the alignment of extensor mechanism, the location of chondral defect, the presence or

not of chondromalacia, and the size of the defect, a treatment algorithm is suggested in Fig. 1.

Conclusion

Arthroscopy and MRI have significantly improved the diagnosis of patellar chondral defects, encouraging in this way the treatment of this challenging entity. Traditional

cartilage resurfacing techniques including microfracture, drilling, and abrasion arthroplasty have been used widely so far, but are limited by the inferior wear characteristics of the fibrocartilage tissue that forms the defect. Besides, microfracture technique may compromise subsequent revision surgery due to damage to the subchondral plate. Realignment of extensor mechanism has been associated with good results only in distal and lateral facet lesions. Thus, newer cartilage repair techniques such as osteochondral autograft transfer (mosaic plasty) or periosteal transfer have been introduced, in order to repair the chondral defects with hyaline cartilage. However, studies on the use of mosaic plasty to treat patellar cartilage damage are lacking, and concern over donor site morbidity has limited its routine performance. On the other hand, periosteal transplantation is promising, but the need for periosteum harvesting from medial tibia metaphysis increases patient's morbidity. More recently used techniques such as autologous chondrocyte implantation performed arthroscopically have significantly reduced patient's morbidity, surgical time, recovery, and complications related to open surgery. But the main disadvantage of this technique remains the high cost and the necessity of performing the surgery in two stages.

The optimal treatment of cartilaginous defects of the patella is still elusive. More prospective studies are needed, in order to identify which techniques are cost-effective especially on a long-term basis.

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